BREATHING AND EXCHANGE OF GASES

90 MINUTES
SPECIALISED PYQ

Q1. Vital capacity of lung is {NEET - 2023}

1.	IRV + ERV + TV	2.	IRV + ERV
3.	IRV + ERV + TV + RV	4.	IRV + ERV + TV - RV

Q2.Select the sequence of steps in respiration. {NEET - 2023}

A.	**Diffusion of gases (O_2 and CO_2) across alveolar membrane.**
B.	**Diffusion of O_2 and CO_2 between blood and tissues**
C.	**Transport of gases by the blood**
D.	**Pulmonary ventilation by which atmospheric air is drawn in and CO_2 rich alveolar air is released out.**
E.	**Utilisation of O_2 by the cells for catabolic reactions and resultant release of CO_2**

Choose the correct answer from the options given below:

1.	**(D), (A), (C), (B), (E)**	**2.**	**(C), (B), (A), (E), (D)**
3.	**(B), (C), (E), (D), (A)**	**4.**	**(A), (C), (B), (E), (D)**

Q3. Under normal physiological conditions in human being every 100 ml of oxygenated blood can deliver _____________ml of O_2 to the tissues. {NEET - 2022}

1. 10 ml

2. 2 ml

3. 5 ml

4. 4 ml

4. Both Statement I and Statement II are False

Q4. Which of the following is not the function of conducting part of respiratory system?

{NEET - 2022}

1.	Provides surface for diffusion of O_2 and CO_2
2.	It clears inhaled air from foreign particles
3.	Inhaled air is humidified
4.	Temperature of inhaled air is brought to body temperature

Q5.Which of the following disorders represents decrease in respiratory surface due to damaged alveolar walls? {NEET - 2022}
1. Hypocapnia
2. Bronchitis
3. Asthma
4. Emphysema

Q6. In the regulation of respiration, a chemosensitive area adjacent to the rhythm centre in the medulla region of the brain, is highly sensitive to: {NEET - 2022}

1. HCO3

2. CO2

3. O2

4. N2

Q7. Which of the following statements are correct with respect to vital capacity? {NEET - 2022}

(a)	It includes ERV, TV and IRV
(b)	Total volume of air a person can inspire after a normal expiration
(c)	The maximum volume of air a person can breathe in after forced expiration
(d)	It includes ERV, RV and IRV.
(e)	The maximum volume of air a person can breathe out after a forced inspiration.

Choose the most appropriate answer from the options given below:
1. (b), (d) and (e)
2. (a), (c) and (d)
3. (a), (c) and (e)
4. (a) and (e)

Q8. Identify the region of human brain which has pneumotaxic centre that alters respiratory rate by reducing the duration of inspiration. {NEET - 2022}

1.	Medulla	2.	Pons
3.	Thalamus	4.	Cerebrum

Q9. The partial pressures (in mm Hg) of oxygen (O_2) and carbon dioxide (CO_2) at alveoli (the site of diffusion) are : **{NEET - 2021}**

1. pO_2=95 and pCO_2 = 40

2. pO_2 = 159 and pCO_2 =0.3

3. pO_2 = 104 and pCO_2=40

4. pO_2 = 40 and pCO_2=45

Q10.Select the favorable conditions required for the formation of oxyhemoglobin at the alveoli. **{NEET - 2021}**

1. High pO_2, high pCO_2, less H^+, higher temperature

2. Low pO_2, low pCO_2, more H^+, higher temperature

3. High pO_2, low pCO_2, less H^+, lower temperature

4. Low pO_2 high pCO_2 more H^+, higher temperature

Q11. Select the correct events that occur during inspiration. **{NEET - 2020}**

a. Contraction of diaphragm

b Contraction of external inter-costal muscles

c. Pulmonary volume decreases

d. Intra pulmonary pressure increases

1. (c) and (d)

2. (a), (b) and (d)

3. only (d)

4. (a) and (b)

Q12. Identify the wrong statement with reference to transport of oxygen: **{NEET - 2020}**

1.	Partial pressure of CO_2 can interfere with O_2 binding with haemoglobin.
2.	Higher H^+ concentration in alveoli favours the formation of oxyhaemoglobin.
3.	Low pCO_2 in alveoli favours the formation of oxyhaemoglobin.
4.	Binding of oxygen with haemoglobin is mainly related to partial pressure of O_2.

Q13. The Total Lung Capacity (TLC) is the total volume of air accommodated in the lungs at the end of forced inspiration. {NEET - 2020}

This includes:

1.	**RV; IC (Inspiratory Capacity); EC (Expiratory Capacity); and ERV**
2.	**RV; ERV; IC and EC**
3.	**RV; ERV; VC (Vital Capacity) and FRC (Functional Residual Capacity)**
4.	**RV (Residual Volume); ERV (Expiratory Reserve Volume); TV (Tidal Volume); and IRV (Inspiratory Reserve Volume)**

Q14. Match the following columns and select the correct option: {NEET - 2020}

Column I		Column II	
(a)	Pneumotaxic Centre	(i)	Alveoli
(b)	O 2 Dissociation curve	(ii)	Pons region of the brain
(c)	Carbonic Anhydrase	(iii)	Haemoglobin
(d)	Primary site of exchange of gases	(iv)	R.B.C.

Options:	(a)	(b)	(c)	(d)
1.	(i)	(iii)	(ii)	(iv)
2.	(ii)	(iii)	(iv)	(i)
3.	(iii)	(ii)	(iv)	(i)
4.	(iv)	(i)	(iii)	(ii)

Q15. Tidal Volume and Expiratory Reserve Volume of an athlete is 500 mL and 1000 mL respectively. What will be his Expiratory Capacity if the Residual Volume is 1200 mL?

{NEET - 2019}

1.	2700 mL	2.	1500 mL
3.	1700 mL	4.	2200 mL

Q16. Due to increasing air-borne allergens and pollutants, many people in urban areas are suffering from respiratory disorder causing wheezing due to: **{NEET - 2019}**

1.	reduction in the secretion of surfactants by pneumocytes
2.	benign growth on mucous lining of nasal cavity
3.	inflammation of bronchi and bronchioles
4.	proliferation of fibrous tissues and damage of the alveolar walls

Q17. Select the correct statement: {NEET - 2019}

1.	Expiration occurs due to external intercostal muscles.
2.	Intrapulmonary pressure is lower than atmospheric pressure during inspiration.
3.	Inspiration occurs when atmospheric pressure is less than intrapulmonary pressure.
4.	Expiration is initiated due to the contraction of the diaphragm.

Q18. The maximum volume of air a person can breathe in after a forced expiration is known as:

{NEET - 2019}

1. Expiratory Capacity

2. Vital Capacity

3. Inspiratory Capacity

4. Total Lung Capacity

Q19. Which of the following options correctly represents the lung conditions in asthma and emphysema, respectively? {NEET - 2018}

1.	Inflammation of bronchioles; decreased respiratory surface
2.	Increased number of bronchioles; increased respiratory surface
3.	Increased respiratory surface; inflammation of bronchioles
4.	Decreased respiratory surface; inflammation of bronchioles

Q20. Match the items given Column I with those in Column II and select the correct option given below: {NEET - 2018}

Column I		Column II	
(a)	Tidal volume	(i)	2500-3000 mL

(b)	Inspiratory Reserve	(ii)	1100-1200 mL volume
(c)	Expiratory Reserve	(iii)	500-550 mL volume
(d)	Residual volume	(iv)	1000-1100 mL

Options:	(a)	(b)	(c)	(d)
1.	iii	ii	i	iv
2.	iii	i	iv	ii
3.	i	iv	ii	iii
4.	iv	iii	ii	i

Q21. Which of the following is an occupational respiratory disorder? **{NEET - 2018}**

1. Anthracis

2. Silicosis

3. Botulism

4. Emphysema

Q22. Lungs are made up of air-filled sacs, the alveoli. They do not collapse even after forceful expiration, because of **{NEET - 2017}**

1. Residual Volume (RV)

2. Inspiratory Reserve Volume (IRV)

3. Tidal Volume (TV)

4. Expiratory Reserve Volume (ERV)

Q23.Name the chronic respiratory disorder caused mainly by cigarette smoking

{NEET - 2016}

1. asthma

2. respiratofy acidosis

3. respiratory alkalosis

4. emphysema

Q24.The partial pressure of oxygen in the alveoli of the lungs is: **{NEET - 2016}**

1. equal to that in the blood

2. more than that in the blood

3. less than that in the blood

4. less than that of carbon dioxide

Q25. Lungs do not collapse between breaths and some air always remains in the lungs which can never be expelled because? **{NEET - 2016}**

1.	there is a negative pressure in the lungs
2.	there is a negative intrapleural pressure pulling at the lung walls
3.	there is a positive intrapleural pressure
4.	pressure in the lungs is higher than the atmospheric pressure

Q26. Reduction in pH of blood will? **{NEET - 2016}**

1. reduce the blood supply to the brain

2. decrease the affinity of hemoglobin with oxygen

3. release bicarbonate ions by the liver

4. reduce the rate of heartbeat

Q27. Asthma may be attributed to? **{NEET - 2016}**

1. allergic reaction of the mast cells in the lungs
2. inflammation of the trachea
3. accumulation of fluid in the lungs
4. bacterial infection of the lungs

Q28. Name the chronic respiratory disorder caused mainly by cigarette smoking? **{NEET - 2016}**

1. asthma

2. respiratory acidosis

3. respiratory alkalosis

4. emphysema

Q29. Name the pulmonary disease in which alveolar surface area involved in gas exchange is drastically reduced due to damage in the alveolar walls. **{NEET - 2015}**

1. Pleurisky

2. Emphysema

3. Pneumonia

4. Asthma

Q30. When you hold your breath, which of the following gas changes in blood would first lead to the urge to breathe? **{NEET - 2015}**

1. Falling O_2 concentration

2. Rising CO_2 concentration

3. Falling CO_2 concentration

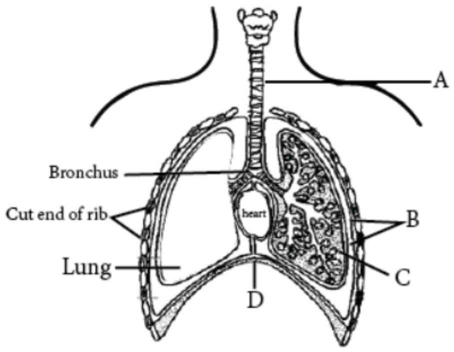
4. Rising CO_2 and falling O_2 concentration

Q31. Approximately seventy percent of carbon dioxide absorbed by the blood will be transported to the lungs: **{AIPMT - 2014}**

1.	as bicarbonate ions

2.	in the form of dissolved gas molecules
3.	by binding to R.B.C
4.	as carbamino-haemoglobin

Q32.The figure shows a diagrammatic view of the human respiratory system with labels A, B, C, and D. Select the option that gives the correct identification, main function, and/or characteristics: **{AIPMT - 2013}**



1.	**B-pleural membrane- surrounds ribs on both sides to provide cushion against rubbing.**
2.	**C-alveoli-thin walled vascular bag-like structures for the exchange of gases.**
3.	**D-lower end of the lungs-diaphragm pulls it down during inspiration.**
4.	**A trachea-long tube supported by complete cartilaginous rings for conducting inspired air.**

Q33.Which one of the following is the correct statement for respiration in humans?

{AIPMT - 2012}

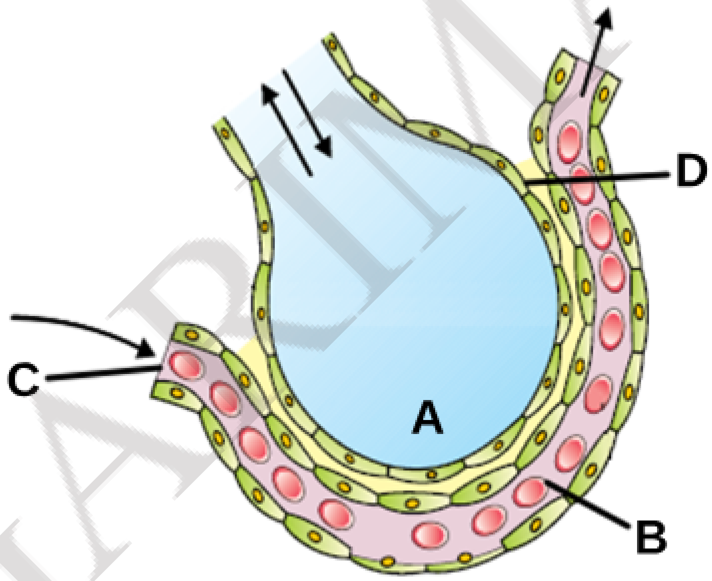
1.	Cigarette smoking may lead to inflammation of the bronchi
2.	Neural signals from the pneumotaxic centre in the pons region of the brain can increase the duration of inspiration

3.	Workers in grinding and stone-breaking industries may suffer from lung fibrosis
4.	About 90% of carbon dioxide (CO_2) is carried by haemoglobin as carbaminohaemoglobin

Q34. **Two friends are eating together at a dining table. One of them suddenly starts coughing while swallowing some food. This coughing would have been due to improper movement of:** **{AIPMT - 2011}**

1.	**diaphragm**	**2.**	**neck**
3.	**tongue**	**4.**	**epiglottis**

Q35. The figure given below shows a small part of the human lung where exchange of gases takes place. In which one of the options given below, the one part A, B, C or D is correctly identified along with its function? **{AIPMT - 2011}**



1.	**A – Alveolar cavity – main site of exchange of respiratory gases**
2.	**D – Capillary wall – exchange of O_2 and CO_2 takes place here**

3.	B – Red blood cell – transport of CO_2 mainly.
4.	C – Arterial capillary – passes oxygen to tissues

ANSWER KEY

1.	(1)	11.	(4)	21.	(2)	31.	(1)
2.	(1)	12.	(2)	22.	(1)	32.	(2)
3.	(3)	13.	(4)	23.	(4)	33.	(3)
4.	(1)	14.	(2)	24.	(2)	34.	(4)
5.	(4)	15.	(2)	25.	(2)	35.	(1)
6.	(2)	16.	(3)	26.	(2)		
7.	(3)	17.	(2)	27.	(1)		
8.	(2)	18.	(2)	28.	(4)		
9.	(3)	19.	(1)	29.	(2)		
10.	(3)	20.	(2)	30.	(2)		